

Decodelabs

Data Analytics Internship

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Project Title: Data Cleaning and Preparation

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Abstract.

The objective of this project is to clean and prepare a retail sales data set in Python with the help of Pandas library. The dataset included customer order information such as products, payment methods, shipping information, and sales values. The primary goal of the project is to enhance the dataset's quality and consistency, identifying and resolving missing data, eliminating duplicate entries, and normalizing date formats.

The data was cleaned using various methods to ensure its accuracy and completeness. Missing values in the CouponCode column were identified and handled appropriately, duplicate records were identified and deleted, and the date format was standardized. The cleaned data was then exported to an Excel file to be used for further analysis.

The project illustrates the significance of data preprocessing in the realm of data analytics and showcases the capabilities of Python and Pandas for effectively executing data cleaning tasks efficiently.

Acknowledgement.

I would like to thank the institution for the opportunity and guidance provided to successfully complete this Data Analytics project.

Finally, I would like to express my gratitude to the developers and contributors of Python, Pandas, and OpenPyXL for creating powerful tools and libraries that made this project possible.

Last but not least, I would like to thank all the learning resources and documentation that provided me with an understanding of the concepts of data cleaning and data preprocessing in the completion of this project.

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1. Introduction.

The project is about cleaning and preparing the retail sales data for data analytics. Raw data may include missing data, duplicate entries, and inconsistencies in data formatting that can impact the accuracy of analysis. The aim of this project was to enhance the dataset using Python and the Pandas library.

The data included customer order information, product details, payment information, shipping details and total sales values.

2. Objectives.

The main objectives of this project were:

- Identify missing values in the dataset
- Handle incomplete data appropriately
- Detect and remove duplicate records
- Standardize the date format
- Validate unique order identifiers
- Export a cleaned dataset for further analysis

The dataset contained information related to customer orders, products, payment methods, shipping details, and total sales values.

3. Tools and Technologies used.

Tool / Technology	Purpose
Python	Programming language
Pandas	Data cleaning and manipulation
OpenPyXL	Excel file handling
Visual Studio Code / PowerShell	Running Python scripts
Microsoft Excel	Viewing cleaned dataset

4. Dataset Description.

The dataset used in this project contained the following columns:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

The dataset represented customer purchase transactions from an online retail environment.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAdd	PaymentMet	OrderStatu	TrackingNum	ItemsInCar	CouponCod	ReferralSou	TotalPrice
ORD2000; #####	C72649	Monitor	5	570.62	928 Main	Debit Card	Shipped	TRK37947	7	SAVE10	Instagram		2853.1
ORD2000; #####	C75739	Phone	2	151.35	823 Main	Online	Shipped	TRK91186	3	SAVE10	Referral		302.7
ORD2000; #####	C81728	Tablet	5	550.68	512 Main	Credit Card	Cancelled	TRK42903	8	FREESHIP	Email		2753.4
ORD2000; #####	C33540	Chair	1	273.19	275 Main	Debit Card	Returned	TRK62788	5	SAVE10	Facebook		273.19
ORD2000; #####	C81840	Printer	4	626.01	668 Main	Online	Delivered	TRK29241	8	SAVE10	Email		2504.04
ORD2000; #####	C37249	Phone	2	245.86	934 Main	Credit Card	Shipped	TRK72976	4	SAVE10	Instagram		491.72
ORD2000; #####	C83492	Laptop	1	664.42	986 Main	Gift Card	Returned	TRK96417	6	SAVE10	Facebook		664.42
ORD2000; #####	C41460	Monitor	5	149.55	706 Main	Cash	Shipped	TRK78809	9	FREESHIP	Facebook		747.75
ORD2000; #####	C26817	Phone	2	134.28	904 Main	Gift Card	Cancelled	TRK61042	2		Email		268.56
ORD2000; #####	C31946	Desk	4	509.38	102 Main	Credit Card	Shipped	TRK33478	6	SAVE10	Google		2037.52
ORD2000; #####	C43443	Tablet	5	625.97	333 Main	Credit Card	Returned	TRK98859	9	WINTER15	Instagram		3129.85
ORD2000; #####	C93861	Monitor	3	49.14	831 Main	Online	Returned	TRK48234	7	SAVE10	Email		147.42
ORD2000; #####	C38785	Monitor	2	180.5	179 Main	Debit Card	Pending	TRK20419	6	FREESHIP	Referral		361
ORD2000; #####	C88348	Laptop	3	201.49	903 Main	Credit Card	Returned	TRK81913	4	FREESHIP	Facebook		604.47
ORD2000; #####	C98474	Tablet	2	393.33	980 Main	Debit Card	Pending	TRK79186	3	SAVE10	Instagram		786.66
ORD2000; #####	C39416	Printer	1	473.96	942 Main	Cash	Delivered	TRK54930	3		Google		473.96
ORD2000; #####	C28976	Printer	2	533.81	914 Main	Credit Card	Pending	TRK13194	3	FREESHIP	Email		1067.62
ORD2000; #####	C60672	Tablet	1	423.4	382 Main	Debit Card	Shipped	TRK11972	3		Referral		423.4
ORD2000; #####	C84329	Desk	1	431.44	891 Main	Gift Card	Pending	TRK65191	3	SAVE10	Facebook		431.44
ORD2000; #####	C14552	Monitor	5	224	273 Main	Cash	Returned	TRK59101	9	SAVE10	Google		1120
ORD2000; #####	C54829	Chair	5	62.51	587 Main	Cash	Pending	TRK16718	9	WINTER15	Instagram		312.55
ORD2000; #####	C99023	Monitor	4	342.95	569 Main	Cash	Delivered	TRK52646	6		Google		1371.8
ORD2000; #####	C34416	Monitor	2	49.62	633 Main	Online	Pending	TRK75937	2	SAVE10	Google		99.24
ORD2000; #####	C42372	Phone	4	273.25	755 Main	Debit Card	Cancelled	TRK36168	5	FREESHIP	Referral		1093
ORD2000; #####	C17470	Laptop	2	230.95	956 Main	Online	Delivered	TRK28775	3	WINTER15	Referral		461.9
ORD2000; #####	C42577	Monitor	1	399.98	709 Main	Debit Card	Delivered	TRK69423	3	WINTER15	Email		399.98
ORD2000; #####	C21944	Laptop	5	59.55	296 Main	Online	Delivered	TRK80276	6		Referral		297.75
ORD2000; #####	C97593	Printer	1	472.74	881 Main	Online	Shipped	TRK89117	6	WINTER15	Google		472.74
ORD2000; #####	C59823	Phone	4	160.07	568 Main	Online	Pending	TRK65895	5		Email		640.28
ORD2000; #####	C33260	Desk	3	423.53	930 Main	Credit Card	Delivered	TRK59744	6	SAVE10	Referral		1270.59

Figure 1: Original dataset before data cleaning.

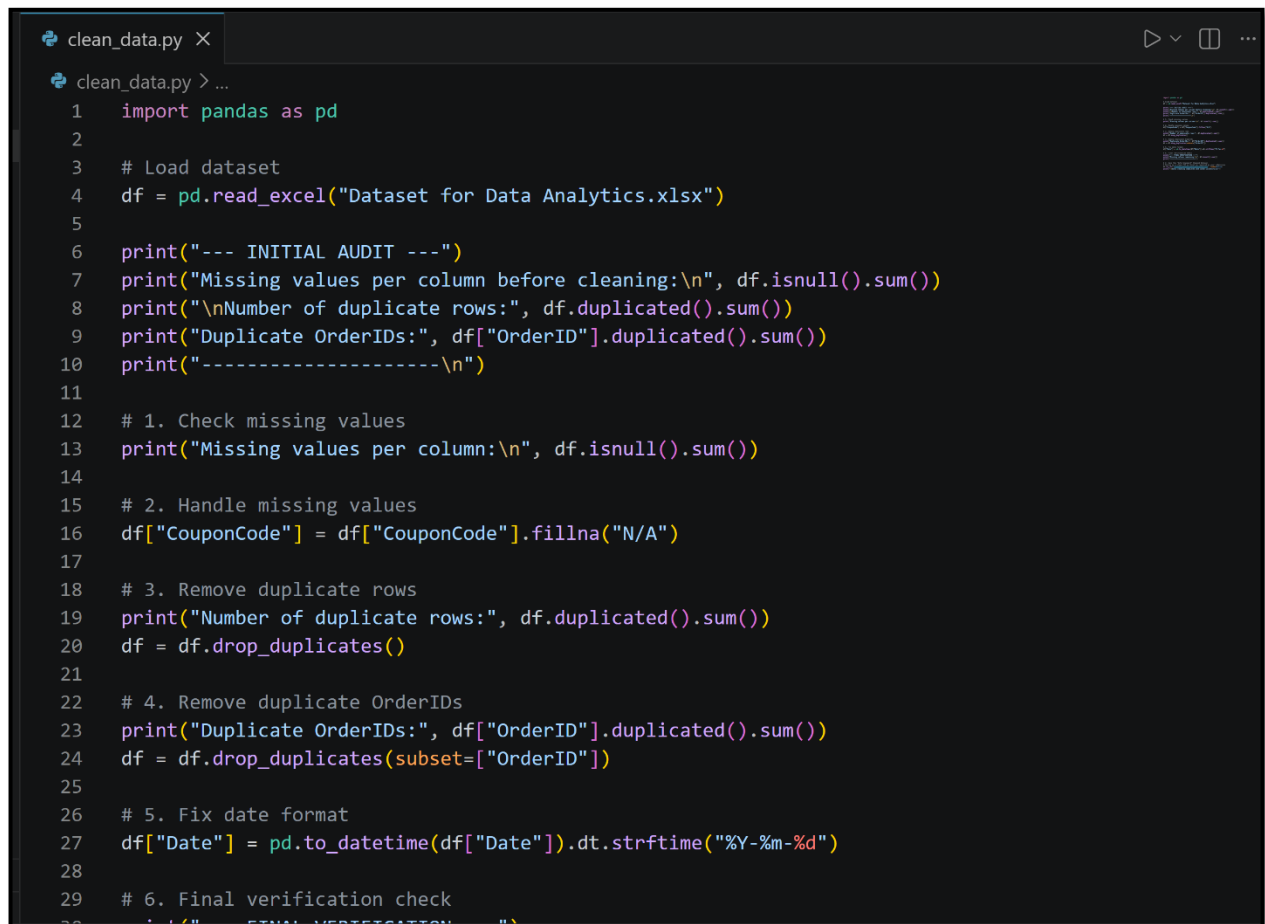
5. Data Cleaning Process.

5.1. Data Cleaning Process.

The dataset was checked for null or missing values using:

df.isnull().sum()

- The CouponCode contained missing values

A screenshot of a Jupyter Notebook interface with a dark theme. The notebook is titled 'clean_data.py'. The code is written in Python and includes comments for each step of the data cleaning process. The steps are: 1. Check missing values using df.isnull().sum(). 2. Handle missing values by filling them with 'N/A' using df['CouponCode'].fillna(). 3. Remove duplicate rows using df.drop_duplicates(). 4. Remove duplicate OrderIDs using df.drop_duplicates(subset=['OrderID']). 5. Fix date format using pd.to_datetime().dt.strftime(). 6. Final verification check. The code is as follows:

```
clean_data.py X
clean_data.py > ...
1  import pandas as pd
2
3  # Load dataset
4  df = pd.read_excel("Dataset for Data Analytics.xlsx")
5
6  print("--- INITIAL AUDIT ---")
7  print("Missing values per column before cleaning:\n", df.isnull().sum())
8  print("\nNumber of duplicate rows:", df.duplicated().sum())
9  print("Duplicate OrderIDs:", df["OrderID"].duplicated().sum())
10 print("-----\n")
11
12 # 1. Check missing values
13 print("Missing values per column:\n", df.isnull().sum())
14
15 # 2. Handle missing values
16 df["CouponCode"] = df["CouponCode"].fillna("N/A")
17
18 # 3. Remove duplicate rows
19 print("Number of duplicate rows:", df.duplicated().sum())
20 df = df.drop_duplicates()
21
22 # 4. Remove duplicate OrderIDs
23 print("Duplicate OrderIDs:", df["OrderID"].duplicated().sum())
24 df = df.drop_duplicates(subset=["OrderID"])
25
26 # 5. Fix date format
27 df["Date"] = pd.to_datetime(df["Date"]).dt.strftime("%Y-%m-%d")
28
29 # 6. Final verification check
30 print("FINAL VERIFICATION")
```

Figure 2: Python script for data cleaning

5.2. Handling Missing Values.

Missing values in the CouponCode column were replaced with “N/A” to ensure the consistency of the dataset using the following python code:

```
df["CouponCode"] = df["CouponCode"].fillna("N/A")
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Quantity          0
UnitPrice         0
ShippingAddress    0
PaymentMethod     0
OrderStatus       0
TrackingNumber    0
ItemsInCart       0
CouponCode        309
ReferralSource    0
TotalPrice        0
dtype: int64
Number of duplicate rows: 0
Duplicate OrderIDs: 0
--- FINAL VERIFICATION ---
Missing values remaining:
OrderID          0
```

Figure 3: Missing value detection output.

5.3. Removing Duplicate Rows.

Duplicate rows were identified and removed using:

```
df = df.drop_duplicates()
```

5.4. Removing Duplicate OrderIDs.

The OrderID column was checked and the duplicate OrderIDs were removed to ensure the uniqueness as it acts as the primary key in the dataset.

Duplicate OrderIDs were removed using:

```
df = df.drop_duplicates(subset=["OrderID"])
```

5.5. Formatting Dates..

The Date column was formatted to the standard format [YYYY-MM-DD] to improve consistency and make future analysis easier using:

```
df["Date"] = pd.to_datetime(df["Date"]).dt.strftime("%Y-%m-%d")
```

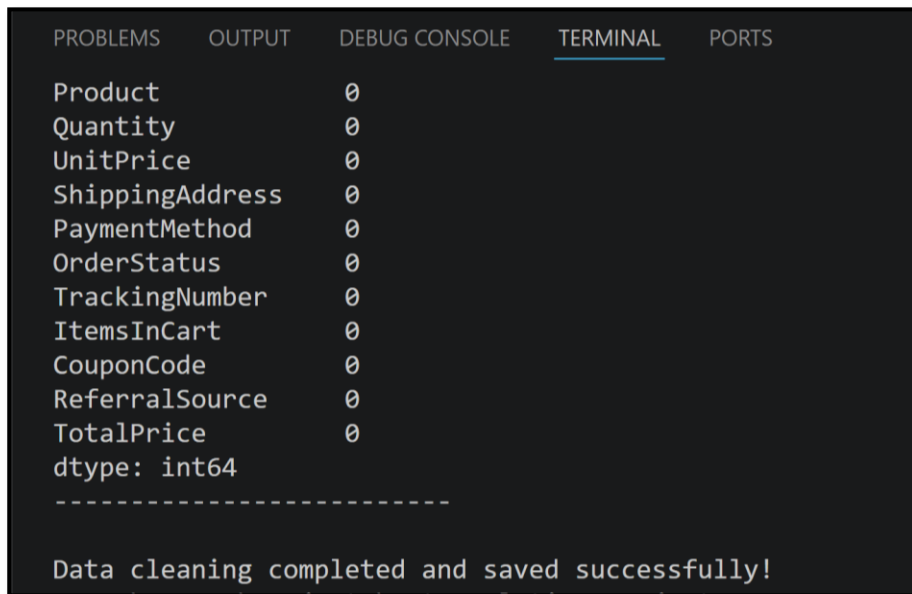
6. Final Results.

After completing the cleaning process:

- All missing values were handled successfully
- No duplicate rows remained
- No duplicate OrderID values existed
- Dates were standardized
- The cleaned dataset was exported successfully

The cleaned dataset was saved as:

Cleaned_Dataset_Data_Analytics.csv



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Product      0
Quantity     0
UnitPrice    0
ShippingAddress 0
PaymentMethod 0
OrderStatus  0
TrackingNumber 0
ItemsInCart  0
CouponCode   0
ReferralSource 0
TotalPrice   0
dtype: int64
-----

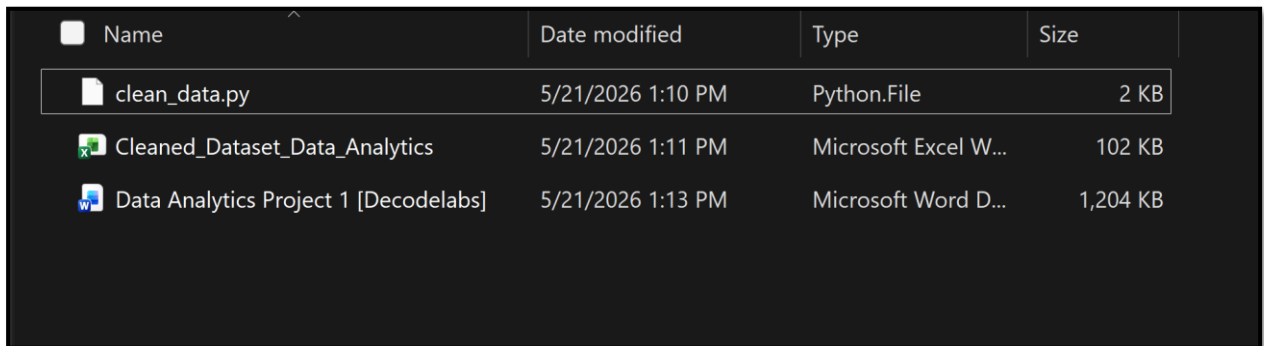
Data cleaning completed and saved successfully!
```

Figure 4: Successful execution of data cleaning script.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	N/A	Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	N/A	Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3	N/A	Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	N/A	Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	N/A	Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
ORD200028	2023-11-21	C59823	Phone	4	160.07	568 Main St	Online	Pending	TRK65895539	5	N/A	Email	640.28
ORD200029	2024-09-20	C33260	Desk	3	423.53	930 Main St	Credit Card	Delivered	TRK59744090	6	SAVE10	Referral	1270.59
ORD200030	2025-03-22	C44267	Monitor	2	426.7	895 Main St	Online	Pending	TRK54751444	5	SAVE10	Referral	853.4
ORD200031	2023-06-30	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	N/A	Facebook	1500.24
ORD200032	2023-05-01	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	N/A	Email	2683.6
ORD200033	2023-06-05	C52184	Chair	1	162.77	912 Main St	Online	Cancelled	TRK73312471	4	WINTER15	Facebook	162.77

Figure 5: Cleaned dataset after preprocessing.

7. Folder Structure.

A screenshot of a file explorer window with a dark theme. It displays a list of files and folders. The first item is a Python file named 'clean_data.py'. The second item is a folder named 'Cleaned_Dataset_Data_Analytics'. The third item is a Word document named 'Data Analytics Project 1 [Decodelabs]'.

 Name	Date modified	Type	Size
 clean_data.py	5/21/2026 1:10 PM	Python.File	2 KB
 Cleaned_Dataset_Data_Analytics	5/21/2026 1:11 PM	Microsoft Excel W...	102 KB
 Data Analytics Project 1 [Decodelabs]	5/21/2026 1:13 PM	Microsoft Word D...	1,204 KB

Figure 6: Final project folder structure.

8. Exploratory Data Analysis.

After completing the data cleaning process, exploratory data analysis was performed to understand patterns and trends within the dataset.

Several statistical techniques and visualizations were used to analyze customer orders, revenue, payment methods, and product performance.

Key Findings

- The dataset contained 1200 customer order records.
- Total revenue generated from all orders was 1,264,761.96.
- The average order value was approximately 1053.97.
- The most frequently used payment method was Online payment.
- Printers and Tablets were among the highest-selling products.
- Some products generated significantly higher revenue compared to others.

A bar chart visualization was created to represent the top-performing products based on total sales revenue.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Degree\Projects\DataAnalytics Project> python eda_analysis.py

--- DATASET OVERVIEW ---
  OrderID  Date  CustomerID  Product  ...  ItemsInCart  CouponCode  ReferralSource  TotalPrice
0  ORD200000  2023-01-04  C72649  Monitor  ...  7  SAVE10  Instagram  2853.10
1  ORD200001  2024-08-23  C75739  Phone  ...  3  SAVE10  Referral  302.70
2  ORD200002  2024-02-27  C31728  Tablet  ...  8  FREESHIP  Email  2753.40
3  ORD200003  2023-10-15  C33540  Chair  ...  5  SAVE10  Facebook  273.10
4  ORD200004  2025-05-08  C81840  Printer  ...  8  SAVE10  Email  2504.04

[5 rows x 14 columns]

--- DATASET INFO ---
<class 'pandas.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   OrderID             1200 non-null  str
1   Date                1200 non-null  str
2   CustomerID          1200 non-null  str
3   Product             1200 non-null  str
4   Quantity            1200 non-null  int64
5   UnitPrice           1200 non-null  float64
6   ShippingAddress     1200 non-null  str
7   PaymentMethod       1200 non-null  str
8   OrderStatus         1200 non-null  str
9   TrackingNumber      1200 non-null  str
10  ItemsInCart         1200 non-null  int64
11  CouponCode          891 non-null   str
12  ReferralSource       1200 non-null  str
13  TotalPrice          1200 non-null  float64

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Degree\Projects\DataAnalytics Project> python eda_analysis.py

12 ReferralSource 1200 non-null str
13 TotalPrice     1200 non-null float64
dtypes: float64(2), int64(2), str(10)
memory usage: 131.4 KB

--- BASIC STATISTICS ---
      Quantity  UnitPrice  ItemsInCart  TotalPrice
count  1200.000000  1200.000000  1200.000000  1200.000000
mean    2.945833    356.412750     5.485000    1853.968300
std     1.407557    197.177146     2.281983     819.856558
min     1.000000    11.390000     1.000000     11.390000
25%     2.000000    186.062500     4.000000    410.520000
50%     3.000000    364.210000     5.000000    823.615000
75%     4.000000    521.570000     7.000000    1578.475000
max     5.000000    699.930000    10.000000    3456.400000

Total Revenue:
1264761.96

Average Total Price:
1853.9683

Median Total Price:
823.615

Most Common Payment Method:
Online

--- ORDER STATUS COUNT ---
OrderStatus
Cancelled    250
Returned    247
Pending     237
Shipped     235
Delivered   231
Name: count, dtype: int64

--- TOP SELLING PRODUCTS ---
Product
Printer     181
Tablet     179
Chair      178
Laptop     173
Desk       170
Monitor    163
Phone      156
Name: count, dtype: int64

--- HIGHEST TOTAL PRICES ---
  OrderID  Date  CustomerID  Product  ...  ItemsInCart  CouponCode  ReferralSource  TotalPrice
789  ORD200789  2023-08-17  C57276  Tablet  ...  10  SAVE10  Email  3456.40
1122  ORD201122  2023-06-07  C38848  Monitor  ...  8  NaN  Facebook  3398.95
632  ORD200632  2023-05-02  C67268  Laptop  ...  7  WINTER15  Facebook  3390.80
460  ORD200460  2023-11-26  C13877  Chair  ...  5  NaN  Facebook  3384.90
328  ORD200328  2023-02-28  C18404  Tablet  ...  7  SAVE10  Google  3370.20

[5 rows x 14 columns]
```

Figure 7: EDA terminal output.

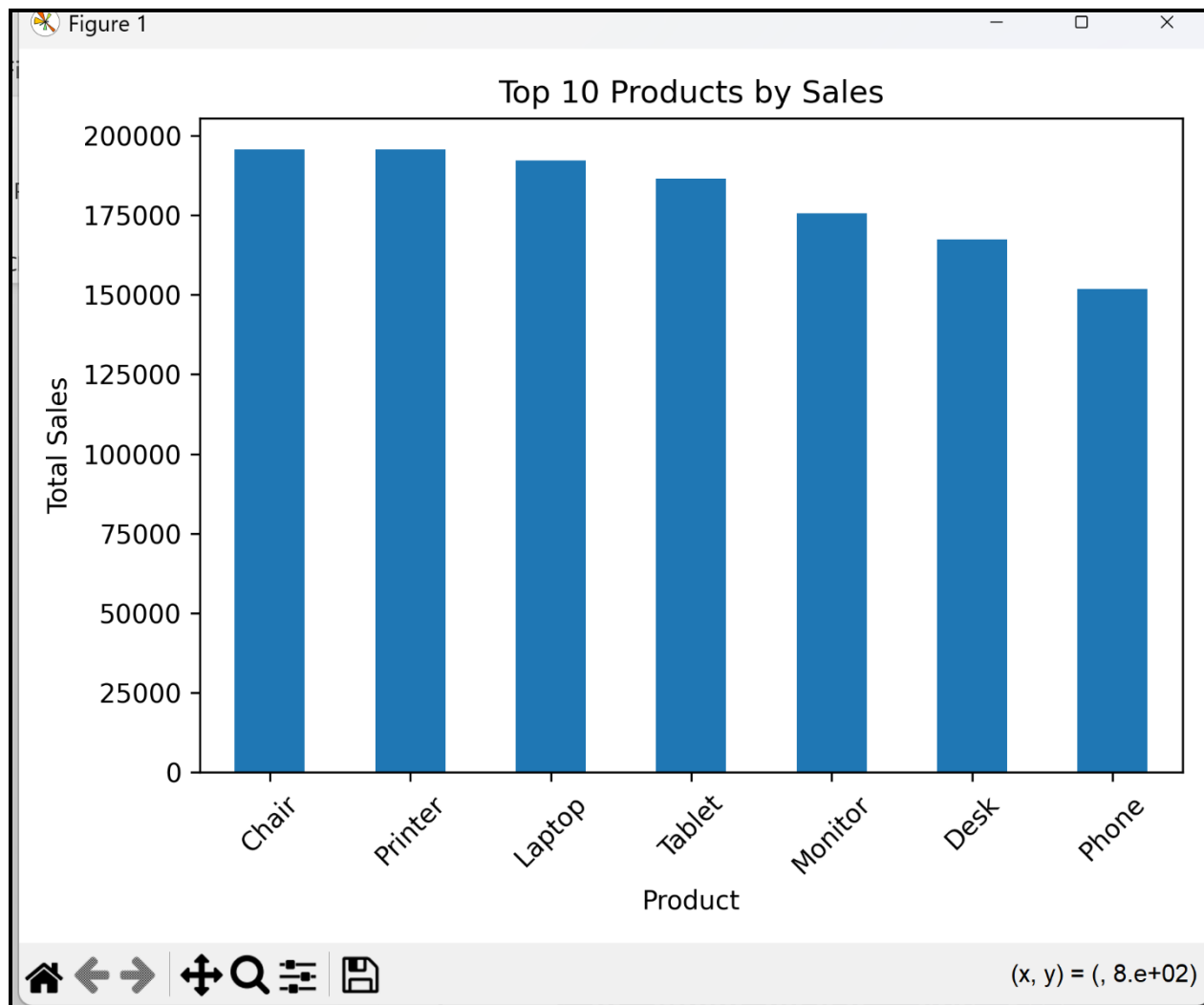


Figure 8: Bar chart graph.

9. SQL Data Analysis.

9.1. SQL Database Creation

The cleaned dataset was imported into an SQLite database using Python. SQL queries were executed to analyze sales performance, payment methods, and product demand.

The cleaned excel dataset was imported into an SQLite database:

sales_data.db

The dataset was stored inside a table called: **sales**

Using the following python code:

```
df.to_sql("sales", conn, if_exists="replace", index=False)
```

9.2. SQL Queries Performed

1. Total Revenue Analysis

The SQL query calculated the total revenue generated from all customer orders.

2. Top Selling Products

A query was executed to identify the products with the highest number of orders.

3. Payment Method Analysis

SQL aggregation queries were used to analyze customer payment preferences.

Key Findings:

- Total revenue generated was 1,264,761.96.
- Printers and Tablets were among the top-selling products.
- Online payment was the most frequently used payment method.

```

clean_data.py  eda_analysis.py  project3_sql.py  Extension: SQLite
project3_sql.py > ...
1  import sqlite3
2  import pandas as pd
3
4  # Load cleaned dataset
5  df = pd.read_excel("Cleaned_Dataset_Data_Analytics.xlsx")
6
7  # Create/connect database
8  conn = sqlite3.connect("sales_data.db")
9
10 # Create SQL table from dataframe
11 df.to_sql("sales", conn, if_exists="replace", index=False)
12
13 # Query 1: Total revenue
14 query1 = """
15 SELECT SUM(TotalPrice) AS Total_Revenue
16 FROM sales;
17 """
18
19 result1 = pd.read_sql(query1, conn)
20
21 print("\n--- TOTAL REVENUE ---")
22 print(result1)
23
24 # Query 2: Top selling products
25 query2 = """
26 SELECT Product, COUNT(*) AS Total_Orders
27 FROM sales
28 GROUP BY Product
29 ORDER BY Total_Orders DESC
30 """
31
32
33 result2 = pd.read_sql(query2, conn)
34
35 print("\n--- TOP SELLING PRODUCTS ---")
36 print(result2)
37
38 # Query 3: Payment methods
39 query3 = """
40 SELECT PaymentMethod, COUNT(*) AS Count
41 FROM sales
42 GROUP BY PaymentMethod;
43 """
44
45 result3 = pd.read_sql(query3, conn)
46
47 print("\n--- PAYMENT METHODS ---")
48 print(result3)
49
50 # Close database connection
51 conn.close()
52
53 print("\nSQL analysis completed successfully!")

```

Figure 9: SQL Code.

```

PS D:\Degree\Projects\DataAnalytics Project> python project3_sql.py

--- TOTAL REVENUE ---
Total_Revenue
0      1264761.96

--- TOP SELLING PRODUCTS ---
Product  Total_Orders
0  Printer           181
1  Tablet            179
2   Chair            178
3  Laptop            173
4   Desk             170

--- PAYMENT METHODS ---
PaymentMethod  Count
0          Cash     246
1  Credit Card     234
2  Debit Card      232
3   Gift Card      230
4        Online     258

SQL analysis completed successfully!
PS D:\Degree\Projects\DataAnalytics Project>

```

Figure 10: SQL Output

10. Conclusion.

This project successfully demonstrated the process of cleaning and preparing raw data using Python and Pandas. Data cleaning is an essential step in the data analytics workflow because accurate analysis depends on high-quality data.

The final cleaned dataset is now ready for further exploration data analysis, visualization, and business insight generation. The project also demonstrated SQL-based analysis and data visualization techniques for business insight generation.

11. References.

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